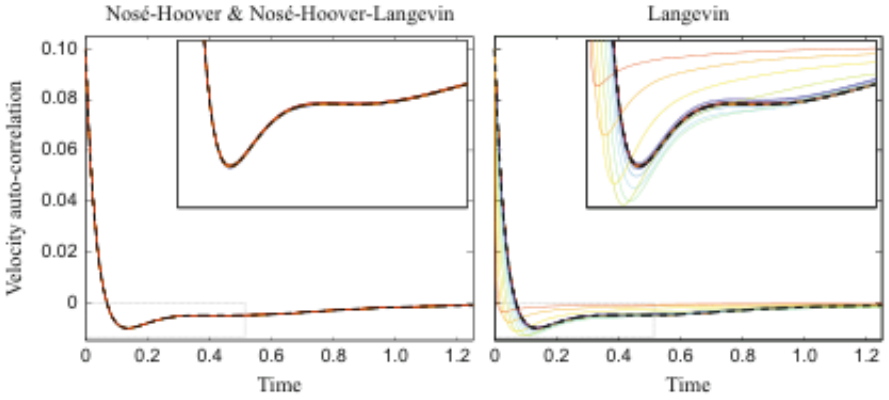


Fig. 8.4 The computed velocity auto-correlation functions are plotted for Nosé-Hoover and Nosé-Hoover-Langevin dynamics (*left*), where we overlay plots from simulations using all possible parameter sets where $\gamma \in \{0, 0.05, 0.5, 5, 50\}$ and $\mu \in \{20, 2, 0.2, 0.02\}$. We additionally plot the same result for Langevin dynamics (*right*), overlaying curves for $\gamma \in \{0.05, 0.1, 0.2, \ldots, 51.2\}$, where high to low friction is plotted from *red* to *blue* respectively. The salient qualitative features of the results remain qualitatively unchanged when using different values of the parameters for the Nosé-Hoover thermostats (see *inset*) whereas the Langevin thermostat has a much harsher effect on the dynamics, with significant differences as the friction is increased

It may seem that the difference between Nosé-Hoover-Langevin and Langevin dynamics is a technical one that essentially hinges on choice of parameters. Both methods are ergodic and it may appear that by simply choosing a small value of friction in Langevin dynamics, we can match the behavior of Nosé-Hoover-Langevin to any desired accuracy. However, this is most definitely not the case, as a careful analysis of Figs. 8.3 and 8.4 reveals. If an accurate velocity autocorrelation function is required, then the friction parameter in Langevin dynamics needs to be chosen very small (much less than $\gamma = 0.1$), see Fig. 8.4. But if the friction is chosen in this way, the convergence to canonical equilibrium is drastically impaired (Figs. 8.2 and 8.3). By contrast it is relatively easy to find parameter choices for Nosé-Hoover-Langevin which provide quite rapid convergence of canonical averages while preserving the accuracy of autocorrelation functions.

The convergence to equilibrium for Nosé-Hoover-Langevin is difficult to calculate analytically. The only existing analysis is that of [225] which restricts consideration to the convergence of the kinetic energy. It turns out that one may obtain a nonlinear system in terms of the following two variables:

$$\bar{\theta}(t) = \int_{\mathbb{D}\times\mathbb{R}} 2K\rho(\boldsymbol{q}, \boldsymbol{p}, \xi, t)\mathrm{d}\omega\mathrm{d}\xi,$$

where K is the kinetic energy, and the mean auxiliary variable ξ

$$\bar{\xi}(t) = \int_{\mathcal{D}\times\mathbb{R}} \xi\rho(\boldsymbol{q},\boldsymbol{p},\xi,t)\mathrm{d}\omega\mathrm{d}\xi.$$

ρ that appears here is the normalized pre-equilibrium density which evolves according to the Fokker-Planck equation of the extended Nosé-Hoover-Langevin system.

It is then possible to perform a local stability analysis to characterize the convergence to equilibrium of $\bar{\theta}$ and $\bar{\xi}$ to the equilibrium $(\mathrm{N_d k_B T}, 0)$. Although the analysis makes certain stringent assumptions they are verified in [225] for a Lennard-Jones molecular system.

8.4.4 Velocity Rescaling

An alternative stochastic-dynamical method for thermostatting molecular dynamics is the stochastic velocity rescaling method proposed by Bussi and collaborators [60, 63] and somewhat generalized in [225]. The equations are (for the case $\boldsymbol{M} = \boldsymbol{I}$, considered for simplicity):

$$\begin{aligned}\mathrm{d}\boldsymbol{q} &= \boldsymbol{p}\mathrm{d}t\\ \mathrm{d}\boldsymbol{p} &= -\nabla U(\boldsymbol{q})\,\mathrm{d}t - \Psi(K)\boldsymbol{p}\,\mathrm{d}t + \sqrt{2k_\mathrm{B}\mathrm{T}\Phi(K)}\boldsymbol{p}\mathrm{d}W,\end{aligned}$$

where $W(t)$ is a single Wiener process, Φ is an arbitrary smooth positive function and Ψ is defined by

$$\Psi(K) \stackrel{\mathrm{def}}{=} (2K - (1+\mathrm{N_d})k_\mathrm{B}T)\Phi(K) - 2k_\mathrm{B}\mathrm{T}K\frac{\mathrm{d}\Phi}{\mathrm{d}K}.$$

With these definitions, the method preserves the canonical probability density ρ_β. We do not prove the ergodicity of the method here, but expect that it would follow by an analysis similar to that given above for Nosé-Hoover-Langevin.

The original proposal of Bussi et al. had

$$\Phi(K) = \frac{\gamma_\mathrm{VR}}{2K}, \quad \text{so that} \quad \Psi(K) = (1 - (\mathrm{N_d}-1)k_\mathrm{B}\mathrm{T}/2K)\,\gamma_\mathrm{VR},$$

where γ_VR is a positive constant.

For the more general case, the required conditions on Φ are as follows:

$K\Phi(K)$ is bounded as $K \to 0$, $\quad$ $\Phi(K)$ grows at most polynomially as $K \to \infty$.